OOC Practicals

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1.INTRODUCTION TO JAVA

Java is class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible. It is intended to let application developers Write Once and Run Anywhere (WORA), meaning that compiled Java code can run on all platforms that support Java without the need for recompilation. Java was developed by James Gosling at Sun Microsystems Inc. in May 1995 and later acquired by Oracle Corporation and is widely used for developing applications for desktop, web, and mobile devices.

2. INTRODUCTION TO ECLIPSE

Eclipse is a free, open-source Integrated Development Environment (IDE) mainly used for writing, compiling, and running programs—especially in Java. It is developed by the Eclipse Foundation and is widely used by students and professional developers.
Eclipse provides a user-friendly interface where I can write code, manage files, and debug errors all in one place. It supports many programming languages like Java, C++, Python, and more through plugins. This means I can extend its features by installing additional tools based on my needs.

Experiment 1:

1.HELLO WORLD

demo.java | demo1.java | Book2.java | saicha29.java | Shoes.java | Shape.java | Squa

```java
public class helloworld{
    public static void main(string[]args) {
        System.out.println("hello,world!");
    }
}
```

Output:

```
Console

Hello, World!
```

2.Program of Comparison Operators

```
package saicha;

public class Test {

    public static void main(String[] args) {

        int a = 10, b = 20;

        System.out.println("a > b = " + (a > b));
        System.out.println("a < b = " + (a < b));
    }
}
```

Output;

```
a > b = false
a < b = true
```

3. Test Program 2 of Comparison Operators.

```
package saicha;

public class Test {
```

```
    public static void main(String[] args) {

        int a = 10, b = 20;

        System.out.println("a <=  b = " + (a <= b));
        System.out.println("a >= b = " + (a >= b));
    }
}
```

Output;

```
a <=  b = true
a >= b = false
```

4. Arithmetic Operators

```
package saicha;

public class Test {

    public static void main(String[] args) {

        int a = 10, b = 20;

        System.out.println("a + b = " + (a + b));
        System.out.println("a - b = " + (a - b));
        System.out.println("a * b = " + (a * b));
        System.out.println("b / a = " + (b / a));
    }
}
```

Output:

```
a + b = 30
a - b = -10
a * b = 200
b / a = 2
```

5. Modules Operator Program

```
package saicha;

public class Test {
   public static void main(String args[]) {

      int a = 10, b = 20, c = 25;

      System.out.println("b % a = " + (b % a));
      System.out.println("c % a = " + (c % a));
   }
}
```

Output:

```
b % a = 0
c % a = 5
```

6. Increment/Decrement

```
package saicha;
   public class Test {
   public static void main(String args[]) {
      int a = 10;
      int d = 25;

      System.out.println("a++ = " + (a++));
      System.out.println("a-- = " + (a--));
```

```
        System.out.println("d++ = " + (d++));
        System.out.println("d-- = " + (d--));
    }
}
```

Output:

```
a++ = 10
a-- = 11
d++ = 25
d-- = 26
```

7. Program to Display text 5 times.

```
package saicha;
class Test {
    public static void main(String[] args) {
        int n = 5;

        // loop from 1 to 5
        for (int i = 1; i <= n; i++) {
            System.out.println("I love Myself");
        }
    }
}
```

Output:

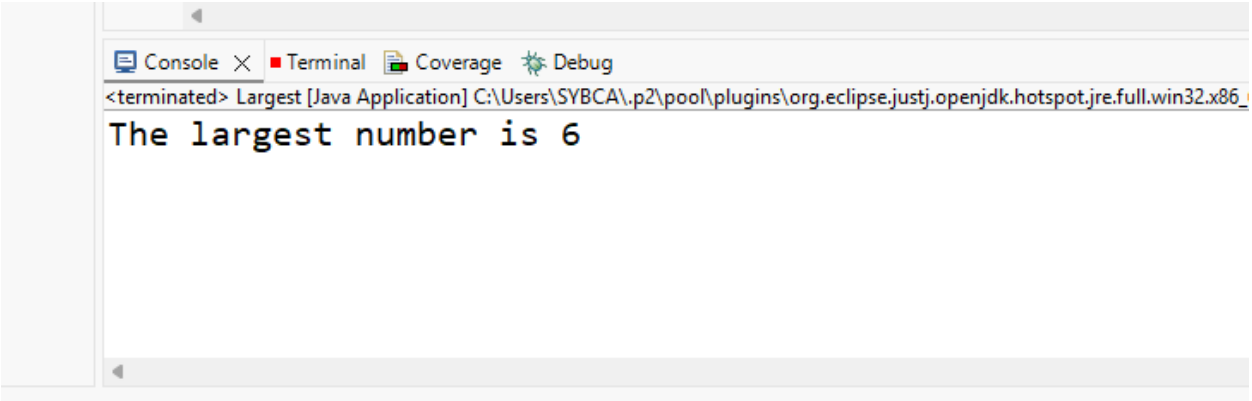
```
I love Myself
I love Myself
I love Myself
I love Myself
I love Myself
```

8. Program to find the largest among three numbers.

```
package project26;
public class Largest {
```

```
  public Largest() {
    // constructor
  }
  public static void main(String[] args) {
    // Program to find the largest among three numbers
    int num1 = 4;
    int num2 = 5;
    int num3 = 6;
    int largest;
    // check the condition
    if (num1 >= num2 && num1 >= num3) {
      largest = num1;
    } else if (num2 >= num1 && num2 >= num3) {
      largest = num2;
    } else {
      largest = num3;
    }
    // display the result
    System.out.println("The largest number is " + largest);
  }
}
```

Output:



9. Java IF Statement

```java
package project26;

public class Geeks {

    public Geeks() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {
                int i = 10;

                if (i < 15) {
                    System.out.println("Condition is True");
                }
            }
        }

    |
```

Output:

```
Console  Terminal  Coverage  Debug
<terminated> Geeks (1) [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.
Condition is True
```

10. Java IF else Statement

```java
package saicha;

public class Geeks {

    public Geeks() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {

                int i = 10;

                if (i < 15) {
                    System.out.println("Condition is True");
                } else {
                    System.out.println("Condition is False");
                }
            }
}
```

Output:

```
Javadoc  Declaration  Console  Coverage
<terminated> Geeks [Java Application] C:\Users\SY BCA\.p2\pool\plugins\org.eclipse.justj.ope
Condition is True
```

11. Java Nested-if Statement

```java
package project26;

public class Geek {

    public Geek() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {

                int i = 10;

                // outer if statement
                if (i < 15) {
                    System.out.println("i is smaller than 15");

                    // nested if statement
                    if (i == 10) {
                        System.out.println("i is exactly 10");
                    }
                }
            }
        }
```

Output:

```
Console  Terminal  Coverage  Debug
<terminated> Geek [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.2.v202
i is smaller than 15
i is exactly 10
```

12. Printing Hello world 5 times

```java
package saicha;
class Geeks {
   public static void main(String args[]) {
```

```
        // writing a for loop
        // to print Hello World 5 times
        for (int i = 1; i <= 5; i++) {
            System.out.println("Hello World");
        }
    }
}
```

Output;

```
Hello World
Hello World
Hello World
Hello World
Hello World
```

13. Factorial Iterative

```java
package saicha;

public class FactorialIterative {

    public FactorialIterative() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {

                int number = 10;
                long factorial = 1;

                if (number < 0) {
                    System.out.println("Factorial of a negative number is not defined.");
                } else {
                    for (int i = 1; i <= number; i++) {
                        factorial *= i;
                    }

                    System.out.println("Factorial of " + number + " is " + factorial);
                }
            }
        }

```

Output:

```
Javadoc  Declaration  Console  Coverage  Eclipse IDE for Java Develo
<terminated> FactorialIterative [Java Application] C:\Users\SY BCA\.p2\pool\plugins\org.eclip
Factorial of 10 is 3628800
```

14. For loop

```java
package saicha;

public class Geeks {

    public Geeks() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {

                // for loop to print numbers from 1 to 5
                for (int i = 1; i <= 5; i++) {
                    System.out.println(i);
                }

                System.out.println("Loop has ended.");
            }
        }
```

Output:

```
Javadoc  Declaration  Console  Coverage  Eclipse IDE
<terminated> Geeks [Java Application] C:\Users\SY BCA\.p2\pool\plugins\org
1
2
3
4
5
Loop has ended.
```

14. For loop (1 to 10)

```java
package saicha;

public class Geeks {

    public Geeks() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {

                // for loop to print numbers from 1 to 5
                for (int i = 1; i <= 10; i++) {
                    System.out.println(i);
                }

                System.out.println("Loop has ended.");
            }
        }
```

Output:

```
Javadoc  Declaration  Console  Coverage
<terminated> Geeks [Java Application] C:\Users\SY BCA\.p2\pool\pl
1
2
3
4
5
6
7
8
9
10
Loop has ended.
```

15. Fibonacci Series till 10 terms.

```java
package saicha;

public class Main {

    public Main() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {

                int n = 10, firstTerm = 0, secondTerm = 1;

                System.out.println("Fibonacci Series till " + n + " terms:");

                for (int i = 1; i <= n; i++) {
                    System.out.print(firstTerm + " ");

                    int nextTerm = firstTerm + secondTerm;
                    firstTerm = secondTerm;
                    secondTerm = nextTerm;
                }
            }
}
```

Output:

```
Javadoc  Declaration  Console  Coverage
<terminated> Main [Java Application] C:\Users\SY BCA\.p2\pool\plu
Fibonacci Series till 10 terms:
0 1 1 2 3 5 8 13 21 34
```

16. Test Student.

```java
package saicha;
class Student {
    int id;
    String name;
}

public class TestStudent {
    public static void main(String args[]) {

        Student s1 = new Student();
        Student s2 = new Student();

        s1.id = 101;
        s1.name = "Sonu";

        s2.id = 102;
        s2.name = "Amit";

        System.out.println(s1.id + " " + s1.name);
        System.out.println(s2.id + " " + s2.name);
    }
}
```

Output:

```
101 Sonu
102 Amit
```

17. Class with Methods and Constructors.

```java
package saicha;

public class Car {

    String brand, color;
    int speed;

    // Constructor
    Car(String brand, String color, int speed) {
        this.brand = brand;
        this.color = color;
        this.speed = speed;
    }

    void displayCar() {
        System.out.println("Brand: " + brand + ", Color: " + color + ", Speed: " + speed);
    }

    public static void main(String[] args) {
        Car car1 = new Car("Toyota", "Red", 180);
        car1.displayCar();
    }
}
```

Output:

```
Javadoc  Declaration  Console ×  Coverage  Eclipse ID
<terminated> Car [Java Application] C:\Users\SY BCA\.p2\pool\plugins\org.
Brand: Toyota, Color: Red, Speed: 180
```

2. Area of square

```java
package project26;
public class SquareAreaDemo2 {
	public static void main(String[] args) {
		double side =4.0;
		double area= side*side;
		System.out.println("Area of Square is:"+area);
	}
}
```

Output:

```
Console ×  Terminal  Coverage  Debug
<terminated> SquareAreaDemo2 [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86
Area of Square is:16.0
```

3.Area Of Triangle

```java
package project26;

public class Areaoftriangle {

    public Areaoftriangle() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {
            double base=10.0;
            double height=5.0;
            double area=(base*height)/2;
            System.out.println("Area of Triangle is:"+area);


    }

}
```

Output:

```
Console  Terminal  Coverage  Debug
<terminated> Areaoftriangle [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.f
Area of Triangle is:25.0
```

4.Area of Rectangle

```java
package project26;

public class areaofrectangle {

    public static void main(String[] args) {
        double length=5.0; double width=5.0;
        double area=length*width;
        System.out.println("Area of Rectangle is:"+area);
    }

}

```

Output:

```
Console  Terminal  Coverage  Debug
<terminated> areaofrectangle [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21
Area of Rectangle is:25.0
```

5. Area of Circle

```java
package project26;

public class Areaofcircle {

    public Areaofcircle() {
        // TODO Auto-generated constructor stub
    }

    public static void main(String[] args) {
        int radius;
        double pi=3.142,area;
        radius=5;
        area=pi*radius*radius;
        System.out.println("Area of circle is:"+ area);


    }

}

```

Output:

```
Console   Terminal   Coverage   Debug
<terminated> Areaofcircle (1) [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre
Area of circle is:78.55
```

Experiment 2:

```java
public class Book {
  // instance variables
  private String title;
```

```java
private String author;
private double price;
private int option;
// constructor
public Book(String title, String author, double price) {
    this.title = title;     // use this keyword
    this.author = author;
    this.price = price;
}
// getters
public String getTitle() {
    return title;
}
public String getAuthor() {
    return author;
}
public double getPrice() {


    return price;
}
public int getOption() {
    return option;
}
// setter for option
public void setOption(int option) {
    this.option = option;
}
// method to update price based on option
public void setPrice(int option) {
    if (option == 1) {
        price = 20.00;
    } else if (option == 2) {
        price = 15.00;
    } else if (option == 3) {
        price = 23.90;
    } else if (option == 4) {
        price = 27.30;
    } else if (option == 5) {
        price = 50.00;
    } else if (option == 6) {
        price = 13.50;
    }
}
// display method
public void display() {
    System.out.println("Title: " + title);
    System.out.println("Author: " + author);
    System.out.println("Price: $" + price);
```

```
  }
  // main method
  public static void main(String[] args) {
      Book b1 = new Book("The Alchemist", "Paulo Coelho", 0);
      b1.setOption(3);
      b1.setPrice(b1.getOption());
      b1.display();
  }
}
```

- //Scanner

```
package saicha;
import java.util.Scanner;
public class TestBook {
  public static void main(String[] args) {
      Scanner input = new Scanner(System.in);
      // Book list
      System.out.println("The Book List");
      System.out.println("1) How to do programming 12th Edition by Liang.");
      System.out.println("2) Malaysian Food Recipe by Nadia Bt Mahmud.");
      System.out.println("3) What is Islam by Ustad Manzoor Malik.");
      System.out.println("4) Urban Legend by Christine R.M.");
      System.out.println("5) Fundamental of Calculus by Prof. Abu.");
      System.out.println("6) How to raise your kids by Salsabila");
      System.out.print("\nChoose your book number to know the price: ");
      int option = input.nextInt();
      // Create book object (dummy values first)
      Book b1 = new Book("Unknown", "Unknown", 0);
      // Set option and price
      b1.setOption(option);
      b1.setPrice(option);
      // Display result
      System.out.println("The price is: $" + b1.getPrice());
      input.close();
  }
}
```

Output:

```
The Book List
1) How to do programming 12th Edition by Liang
2) Malaysian Food Recipe by Nadia Bt Mahmud
3) What is Islam by Ustad Manzoor Malik
4) Urban Legend by Christine R.M.
5) Fundamental of Calculus by Prof. Abu
6) How to raise your kids by Salsabila

Choose your book number to know the price: 3
The price is: 23.9
```

Experiment 3:

```java
package saicha;
public class Dog {
  // Attributes
  String breed;
  String color;
  int age;
  boolean isHungry;
  // Constructor
  public Dog(String breed, String color, int age) {
     this.breed = breed;
     this.color = color;
     this.age = age;
     this.isHungry = true;
  }
  // Method to bark
  public void bark() {
     System.out.println(breed + " dog is barking!");
  }
  // Method to eat
  public void eat() {
     if (isHungry) {
        System.out.println(breed + " dog is eating.");
        isHungry = false;
     } else {
        System.out.println(breed + " dog is not hungry right now.");
```

```java
    }
  }
  // Birthday method
  public void birthday() {
    age++;
    System.out.println(breed + " dog is now " + age + " years old!");
  }
  // Main method
  public static void main(String[] args) {
    Dog dog1 = new Dog("Labrador", "Yellow", 3);
    Dog dog2 = new Dog("German Shepherd", "Black and Tan", 2);
    // Dog 1
    System.out.println("Dog 1: Breed - " + dog1.breed + ", Color - " + dog1.color + ",
Age - " + dog1.age);
    dog1.bark();
    dog1.eat();
    dog1.eat();
    dog1.birthday();
    // Dog 2
    System.out.println("Dog 2: Breed - " + dog2.breed + ", Color - " + dog2.color + ",
Age - " + dog2.age);
    dog2.bark();
    dog2.eat();
    dog2.birthday();
  }
}
```

Output:

```
Maximize <d> Dog [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.1.v20231028-0
Dog 1: Breed - Labrador, Color - Yellow, Age - 3
Labrador dog is barking!
Labrador dog is eating.
Labrador dog is not hungry right now.
Labrador dog is now 4 years old!
Dog 2: Breed - German Shepherd, Color - Black and Tan, Age - 2
German Shepherd dog is barking!
German Shepherd dog is eating.
German Shepherd dog is now 3 years old!
```

Experiment 4:

```java
package saicha;
// Base class
public class Book {
  // private data members
  private int bno;
  private String bname;
  private double price;
  // constructor
  public Book(int bno, String bname, double price) {
      this.bno = bno;
      this.bname = bname;
      this.price = price;
  }
  // display method
  public void display() {
      System.out.println("Book Number: " + bno);
      System.out.println("Book Name: " + bname);
      System.out.println("Price: $" + price);
  }
}
// inherited class
class SpecialEditionBook extends Book {
  private double discount;
  // constructor
  public SpecialEditionBook(int bno, String bname, double price, double discount) {
      super(bno, bname, price);
      this.discount = discount;
  }
  // overriding display method
  public void display() {
      super.display();
      System.out.println("Discount: " + discount + "%");
  }
}
// test class
class BookTest {
  public static void main(String[] args) {
      Book book1 = new Book(101, "The Car", 20.99);
      book1.display();
      System.out.println();
      SpecialEditionBook specialBook1 =
              new SpecialEditionBook(167, "The Alchemist", 56.99, 20.0);
      specialBook1.display();
  }
}
```

Output:

```
Javadoc  Declaration  Console
<terminated> BookTest [Java Application] C:\Users\SY BCA\.p2\pool\plugins\org.e
Book Number: 101
Book Name: The Car
Price: $20.99

Book Number: 167
Book Name: The Alchemist
Price: $56.99
Discount: 20.0%
```

Experiment 5:
one default and one parameterized constructor.

```
package saicha;

public class Bag {

    private double price;
    private String brand;
    private String color;
    private String type; // college / office

    // Default constructor
    public Bag() {
        this.price = 0.0;
        this.brand = "Unknown";
        this.color = "Unknown";
        this.type = "Unknown";
    }

    // Parameterized constructor
    public Bag(double price, String brand, String color, String type) {
        this.price = price;
        this.brand = brand;
        this.color = color;
        this.type = type;
    }

    // display method
    public void display() {
        System.out.println("Price: " + price);
        System.out.println("Brand: " + brand);
        System.out.println("Color: " + color);
```

```
        System.out.println("Type: " + type);
    }
}


class BagTest {
    public static void main(String[] args) {

        // Using default constructor
        Bag bag1 = new Bag();
        System.out.println("Default Bag:");
        bag1.display();

        System.out.println();

        // Using parameterized constructor
        Bag bag2 = new Bag(1500.0, "Skybags", "Black", "College");
        System.out.println("Parameterized Bag:");
        bag2.display();
    }
}
```

Output:

```
Default Bag:
Price: 0.0
Brand: Unknown
Color: Unknown
Type: Unknown

Parameterized Bag:
Price: 1500.0
Brand: Skybags
Color: Black
Type: College
```

Experiment 6:

```java
package saicha;
class Book {
  // Instance variables
  String title;
  String author;
  double price;
  // Default constructor
  public Book() {
      this.title = "Unknown";
      this.author = "Unknown";
      this.price = 0.0;
  }
  // Parameterized constructor (title, author)
  public Book(String title, String author) {
      this.title = title;
```

```
        this.author = author;
        this.price = 0.0;
    }
    // Parameterized constructor (title, author, price)
    public Book(String title, String author, double price) {
        this.title = title;
        this.author = author;
        this.price = price;
    }
}
public class Class {
    public static void main(String[] args) {
        Book book1 = new Book();
        System.out.println("Book1 Title: " + book1.title);
        System.out.println("Book1 Author: " + book1.author);
        System.out.println("Book1 Price: " + book1.price);
        Book book2 = new Book("Amatka", "Karin Tidbeck");
        System.out.println("Book2 Title: " + book2.title);
        System.out.println("Book2 Author: " + book2.author);
        System.out.println("Book2 Price: " + book2.price);
        Book book3 = new Book("Altered Carbon", "Richard K. Morgan", 18.99);
        System.out.println("Book3 Title: " + book3.title);
        System.out.println("Book3 Author: " + book3.author);
        System.out.println("Book3 Price: " + book3.price);
    }
}
```

Output:

```
@ Javadoc  Declaration  Console ×  Coverage  Eclipse
<terminated> Class [Java Application] C:\Users\SY BCA\.p2\pool\plugins\
Book1 Title: Unknown
Book1 Author: Unknown
Book1 Price: 0.0
Book2 Title: Amatka
Book2 Author: Karin Tidbeck
Book2 Price: 0.0
Book3 Title: Altered Carbon
Book3 Author: Richard K. Morgan
Book3 Price: 18.99
```

Experiment 7

```
public class Shoes {
private static final int MIN_SIZE = 1;
private static final int MAX_SIZE = 15;
private String brand;//brand can be replaced with color
```

```java
private double price;
private int size;
public Shoes(String brand, double price, int size) {
  this.brand = brand;
  this.price = price;
  this.size = size;
}
public String getBrand() {
  return brand;
}
public void setBrand(String brand) {
  this.brand = brand;
}
public double getPrice() {
  return price;
}
public void setPrice(double price) {
  if (price < 0) {
      System.out.println("Price Must be greater than zero!\n");
      return;
  }
  this.price = price;
}
public int getSize() {
  return size;
}
public void setSize(int size) {
  if (size > MAX_SIZE && size < MIN_SIZE) {
      System.out.println("Invalid Size!\n");
  }
}
public String toString() {
  return "Shoe [brand = " + brand + ", price = " + price + ", size = " +      size + "]";
}
public static void main(String[] args) {
Shoes myShoes = new Shoes("J.F.", 45.99, 10);
Shoes otherShoes = new Shoes("Addidas", 65.99, 16);
System.out.println("The shoes: ");
System.out.println(myShoes.toString());
System.out.println("Other Shoes: ");
System.out.println(otherShoes.toString());
}
}
```

Output:

```
Console  Problems  Debug Shell  Coverage
<terminated> Shoes [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.2.v20240123-0840\jre\bin\jav
The shoes:
Shoe [brand = J.F., price = 45.99, size = 10]
Other Shoes:
Shoe [brand = Addidas, price = 65.99, size = 16]
```

Experiment 8: Method Overloading in Java

```
package saicha;

class Product {

    // Method overloading
    public int prod(int a, int b, int c) {
        return a * b * c;
    }

    public double prod(double a, double b, double c) {
        return a * b * c;
    }
}

public class Products {
    public static void main(String[] args) {

        Product p = new Product();

        System.out.println(p.prod(1, 2, 3));
        System.out.println(p.prod(1.0, 2.0, 3.0));
    }
}
```

Output:

```
6
6.0
```

Experiment 9: Default parameterized constructor

```
package saicha;

public class Student {

    String name;
    int marks;

    // Default constructor
    Student() {
        this.name = "Unknown";
        this.marks = 0;
    }

    // Parameterized constructor
    Student(String n, int m) {
        this.name = n;
        this.marks = m;
    }

    // Copy constructor
    Student(Student s) {
        this.name = s.name;
        this.marks = s.marks;
    }

    // Display method
    void display() {
        System.out.println("Name: " + name + ", Marks: " + marks);
    }

    public static void main(String[] args) {

        // Default constructor
        Student s1 = new Student();

        // Parameterized constructor
```

```
        Student s2 = new Student("Ravi", 88);

        // Copy constructor
        Student s3 = new Student(s2);

        // Display output
        s1.display();
        s2.display();
        s3.display();
    }
}
```

Output:

```
Name: Unknown, Marks: 0
Name: Ravi, Marks: 88
Name: Ravi, Marks: 88
```

Experiment 10: Exception Handling in Java

1.

```java
class demo {
    public static void main(String[] args) {
        int a = 10;
        int b = 0;

        try {
            int ans = a / b;
            System.out.println("Answer: " + ans);
        } catch (ArithmeticException e) {
            System.out.println("Error: Division by 0!");
        }
    }
}
```

Output:

```
Console  Problems  Debug Shell  Coverage
<terminated> demo [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.
Error: Division by 0!
```

2.

```java
class demo1 {
    public static void main(String[] args){

        int[] numbers = { 1, 2, 3 };
        try {
            // This will throw ArrayIndexOutOfBoundsException
            System.out.println(numbers[5]);

        }
        catch (ArrayIndexOutOfBoundsException e){

            System.out.println("Exception caught: " + e);
        }
        finally{
            System.out.println("This block always executes.");
        }
        System.out.println("Program continues...");
    }
}
```

Output:

```
Console  Terminal  Coverage  Debug
<terminated> demo1 [Java Application] C:\Users\SYBCA\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.2.v20240123-0840\jre\bin\javaw.exe (04-Apr-2026, 10:43:20 am – 10:43:20 am elapsed 0:00:00) [pid: 6044]
Exception caught: java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 3
This block always executes.
Program continues...
```

3.

```java
class saicha29 {
    static void checkAge(int age) { if (age < 18)
 {
        throw new IllegalArgumentException("Age must be 18 or above"); } }
 public static void main(String[] args) { checkAge(15); } }
```

Output:

```
Exception in thread "main" java.lang.IllegalArgumentException: Age must be 18 or above
        at saicha29.checkAge(saicha29.java:2)
        at saicha29.main(saicha29.java:3)
```